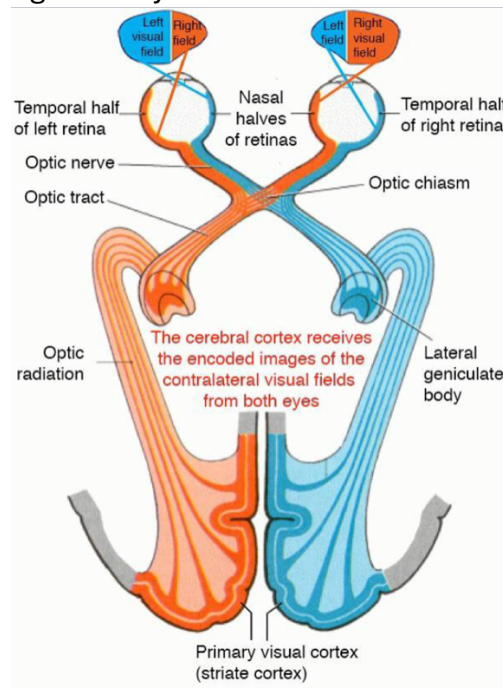


Exam II Study Guide

Lecture 9 Vision

- Stimulus
 - Know the properties of light and its related perceptions
 - Amplitude = brightness = how tall the wave is
 - Wavelength = hue / color = distance from one peak to the next
 - Purity = saturation = how much of the dominant wavelength compared to how much white
- Anatomy of the Eye
 - Know the cranial nerves associated with the eye (related to sensory and motor information)
 - Cranial nerve II = vision
 - Cranial nerve III = eye movements
 - Controls all eye muscles except the superior oblique and the lateral rectus muscles
- Retina
 - The top half of the world hits the bottom of the retina (and vice versa)
 - The left half of the world hits the right side of the retina (and vice versa)
 - Know the organization of the different cell types in the back of the retina
 - Cones are densest at the fovea
 - Know the properties of cones and rods
 - Cones = sharp, central vision and visual acuity / color
 - Rods = vision at low light, low spatial acuity
 - Together, are called photoreceptor cells and encode light first
 - Know the function of bipolar and ganglion cells
 - Bipolar cells get information from rods and cones and pass it to ganglion cells
 - Ganglion cells have axons which form the optic nerve and send the visual information out of the retina
 - First to generate action potentials! Everyone else does graded potentials
 - Understand the meaning of receptive fields and how it changes between cell types
 - Be able to define lateral inhibition and which cell type is responsible for it
 - Lateral inhibition = process where interconnected neurons (i.e. horizontal cells) inhibit their neighbors and produce contrast
- Optic disk
 - Know what it is and its associated properties
 - Optic disk = where optic nerves and blood vessels come together to exit the eye
 - Is a blind spot because there are no photoreceptors
- Optic chiasm

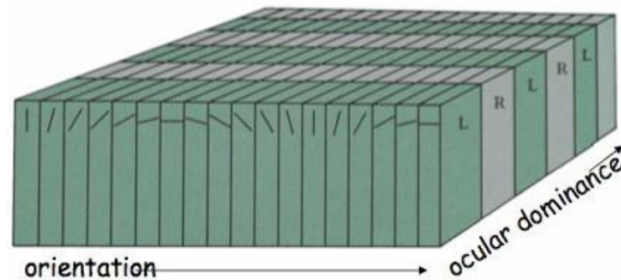
- Know where the information goes when light strikes the right/left visual field and/or right/left eye



- Know what is impaired when certain locations of the visual pathway are damaged

- Lateral geniculate nucleus
 - Know the two cell types and what information they carry
 - Parvocellular neurons = what pathway!
 - Information about visual details, red and green color, etc.
 - Small receptive fields and get their info from midget retinal ganglion neurons around the fovea
 - Magnocellular neurons = where pathway!
 - Information about movement, patterns, contrast between light and dark
 - Large receptive fields, get their info from parasol retinal ganglion neurons located throughout the retina
- Visual cortex
 - Be able to trace the route of visual information from the retina to the cerebral cortex
 - Retina → optic nerve → optic chiasm → lateral geniculate body (thalamus) → primary visual cortex (occipital lobe!)
 - Describe the organization of the primary visual cortex (ocular dominance, orientation columns)
 - Has 6 layers with different cells that connect to different brain areas
 - These layers have ocular dominance (like left or right more) and orientation preferences which form orientation columns

- preference for **neighboring columns** are **not random**:
 - orientation selectivity changes only slightly as you move from column to column and there is a specific and consistent order



- - Ocular dominance = a neuron's response preference for one eye over the other (everything is still split between left and right contralaterally)
- Primary visual cortex: Know its two cell types, what they respond to, and **how the receptive fields change from one cell type to the other**
 - Simple cells = respond to orientation,
 - Complex cells = respond to moving stimuli of particular orientations
- Know the two visual pathways in the visual cortex, their function, and the associated cell types from the retina and LGN
 - Ventral pathway = what = identify objects
 - Retinal cells → midget ganglion neurons
 - LGN cells → parvocellular neurons
 - Dorsal pathway = where/how = identifying guided movements
 - Retinal cells → parasol ganglion neurons
 - LGN cells → magnocellular neurons
- Fusiform gyrus
 - Know where it is located and be able to discuss its potential specialization for facial recognition
 - Located on the basal surface of the occipital and temporal lobes (bottom of brain)
 - Might specialize in facial identification or in general expertise identification

Lecture 10 Motor System

- Muscles of the body
 - Know and name different types of muscles and how they differ from each other
 - Skeletal (striated) muscles = voluntary movement of body in relation to environment
 - Smooth muscles = involuntary movements including the digestive system and other organs
 - Cardiac muscles = heart, has properties from both of the other two muscle types
- Neuromuscular junction (NMJ)

- Know the neurotransmitter used at the NMJ and the sequence of events that happen at a synapse
 - Acetylcholine!
- Proprioception
 - Know the definition of a proprioceptor
 - Proprioceptor = mechanically gated receptors that detect the position or movement of a part of the body
 - Be able to name two proprioceptors and explain their function using an example
 - Muscle spindles = respond to stretch/tension in muscle and cause a contraction (quadricep muscle has these)
 - Golgi tendon organ = respond to increase in muscle tension and cause relaxation (hamstring muscle has these)
- Cranial nerves
 - Know what cranial nerves are associated with carrying or not carrying motor information
 - What cranial nerves carry only motor information
- Pyramidal system
 - Know the function of the primary motor cortex and its association motor areas
 - Is active when someone is intending / planning a movement
 - What brain areas are part of the association motor cortices
 - Know the two types of corticospinal tracts we discussed in class, what body parts
- Extrapyramidal system
 - Name the two brain regions involved in the extrapyramidal system
 - Describe their function
 - Basal ganglia → initiates and plan actions not guided by a specific stimulus, important for spontaneous and self-guided behaviors and makes sure these movements aren't too crazy / vigorous
 - Cerebellum → balance and coordination, tasks that require timing, and aspects of attention (shifting and focusing on visual stimuli)
- Parkinson's disease
 - What are characterizations of Parkinson's disease
 - Muscle tremors, rigidity, slow movement, problems initiating movement
 - What brain regions are involved
 - Substantia nigra and basal ganglia
 - What neurotransmitter is involved
 - Dopamine, less production in substantia nigra due to cell death
 - How is the function of the basal ganglia altered
- What are treatment options for Parkinson's disease and what do they do
 - Pharmacologically, we can try to get more dopamine in the brain

- L-dopa is helpful because it is a precursor to dopamine and can cross the BBB easily, but it does not stop neurons from dying and is ineffective in late stages of the disease
 - Also doesn't work forever
- Surgically, we can do gene therapy, deep brain stimulation, brain transplants, etc.

Lecture 11 Auditory & Somatosensory Systems

Auditory System

- Properties of stimulus (sound)
 - Know the properties of sound and its related perceptions
 - Wavelength → pitch
 - Amplitude → loudness
 - Purity → timbre
- Structure of the ear
 - Know the various structures that make the outer, middle, and inner ear and their associated functions
 - External ear: external auditory canal
 - Locates, collects, and carries sound waves to the eardrum
 - Middle ear: ossicles, tympanic membrane (eardrum)
 - Protects inner ear, converts sound waves into vibrations and amplifies them
 - Inner ear: cochlea, round and oval windows, eustachian tube
 - Transmits vibrations into neural impulses, balance
- Inner ear
 - Know the structure, **properties, and function** of the Cochlea
 - Has a tonotopic organization (highest hertz at base, lowest at middle / apex)
 - Know the structure, properties, and function of the Organ of Corti
 - Converts vibration into neural activity with hair cells
 - Know what hair cells are, where they are located, and how they convert vibrations into a nerve impulse
- Auditory ear pathways
 - Know the events that happen from sound waves entering the ear to being converted into nerve impulses by the inner ear
- Auditory cortex
 - Know the properties of the auditory cortex
- Auditory brain pathway
 - Know the auditory pathways from the cochlea to the primary auditory cortex
- Deafness
 - Know and describe the different types of deafness we talked in class about, their cause, and potential treatment options

Vestibular System

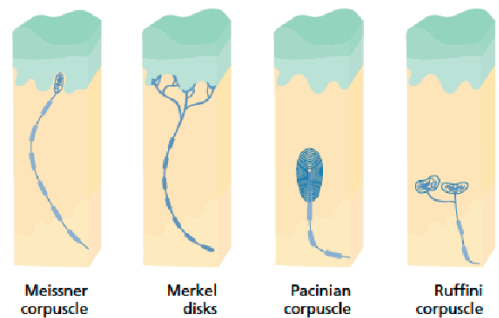
- Structure and function
 - Know the structure and function of the vestibular system
 - Involved in balance and position / movement of head, lets us be aware of our positions, involved in coordination
 - Structure:
 - 3 semicircular canals that each respond to rotational movements
 - Otolith organ responds to gravity
- Physiology
 - Know what the vestibular receptors are, where they are located, and how they are activated
 - Hair cells (stereocilia), partially with otoliths (crystals) on top to increase sensitivity
- Be able to discuss how motion sickness can be explained
 -

Somatosensory system

- Stimulus
 - Know the various stimuli that the somatosensory system responds to
 - Touch, pressure, temperature, tissue damage (pain), body movements and positioning
- Somatosensory receptors
 - Know the various somatosensory receptors and their corresponding function
 - Cutaneous receptors
 - Mechanoreceptors → pressure, touch, texture
 - Thermoreceptors → temperature
 - Nociceptors → pain
 - Proprioceptors → movement
 - Further classification:

Cutaneous receptors can be classified according to their:

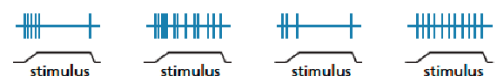
- 1) Structure
- 2) Location



3) Receptive fields



4) Rate of adaptation



- How are these receptors activated?
 - Mechanical stimulation deforms the corpuscle, which stretches and opens sodium channels
- Somatosensory representation
 - Know how somatosensory information is represented in the somatosensory cortex
 - Homunculus!
- Somatosensory pathways
 - Know the pathway from touch and how it differs from the pathway of pain
 - Touch: brainstem / spinal cord → thalamus → cerebral cortex
 - At the thalamus there is also the insula, which is responsible for the emotional aspect of touch
 - Different aspects of body sensation remain separate until they get to the cortex

Lecture 12 Taste and Smell Systems

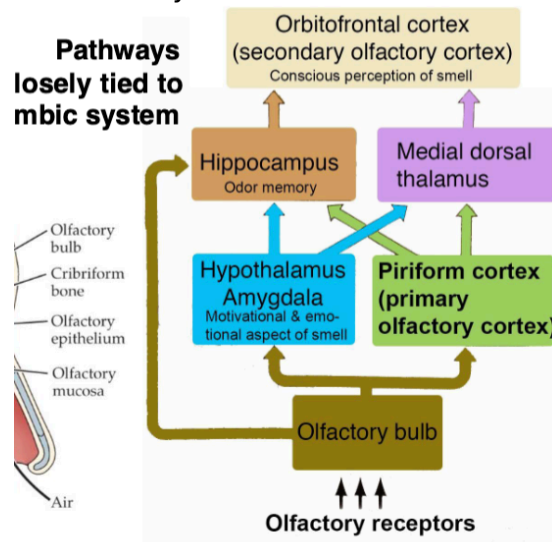
Taste

- Tastes
 - What are the primary tastes for gustation
 - Bitter (toxic compounds)
 - Salty (NaCl, and generally a cation paired with an anion)
 - Sweet (sugars, sometimes with amino acids)
 - Sour (H⁺, a hydrogen cation with an anion)
 - Umami (glutamate)
 - Bitter, sweet, and umami have g-protein couple receptors
 - Sour and salty have ion channel receptors
- Taste buds & receptors
 - What are different types of papillae and where are they located
 - Fungiform papilla → front of tongue
 - Filiform papilla
 - Foliate papilla → side of tongue
 - Circumvallate papilla → back of tongue
 - What are taste receptors and where are they located
 - Cells are modified skin cells located in taste buds
 - Located across the tongue in the different papillae
- Transduction
 - Know how the different taste receptors are activated
 - Have excitatory membranes that release neurotransmitters to excite the neurons around them
- Gustatory pathways
 - Name the three cranial nerves that innervate the tongue and be able to locate them
 - X innervates most of the tongue in a scattered pattern

- VII innervates the anterior (front) 2/3
 - IX innervates the posterior (back) 1/3
 - Know how gustatory information travels to the brain
 - Cranial nerves → solitary nucleus of brainstem → VPM of thalamus → somatosensory cortex and primary gustatory cortex
 - Know which brain structure is the primary gustatory cortex
 - The insula (?)
- Gustation and Olfaction
 - How do both systems work together
 - Work together to create flavor

Olfaction

- Smell
 - What are the primary smells for olfaction
 - Trick question! There are no primaries
- Anatomy of nose
 - Where is the olfactory system located
 - In the nasal cavity
- Olfactory receptors
 - What are olfactory receptors and where are they located
 - Olfactory receptors are located on cilia, which are tiny little hairs
- Transduction
 - Know how olfactory receptors are activated and how they send information to the brain
 - Odorant binds → G protein is activated → Cl⁻ channels open and allow the negative ions to leave, depolarizing the cell
- Olfactory pathways
 - Know how olfactory information travels to the brain



Lecture 13 Wakefulness & Sleep

- Circadian rhythm
 - Define circadian rhythm, give examples
 - Circadian rhythm = an endogenous biological mechanism that occurs on a regular pattern about every 24 hours
 - Sleep patterns, hormone releases, appetite and digestion, etc.
 - How do they differ from other endogenous rhythms?
 - Differ in time (endogenous rhythms can be longer or shorter)
 - What experiments are conducted to show a circadian rhythm exists?
 - Hamster study! When timing of light is shifted, hamster shifts its activity as well
- Biological clock
 - What brain region is the main driver of the circadian rhythm/our biological clock?
 - The suprachiasmatic nucleus (SCN)
 - What experiments prove that that brain region is indeed the main driver?
 - Lesions to the SCN in animals lead to them losing the rhythm in continuous dim light
 - What are the underlying mechanisms (know the different proteins involved and how they are up- or down-regulated at night/daytime, etc.)
 - Upregulation of PER and TIM proteins leads to sleep
 - How can light entrain the biological clock/the brain region?
 - What hormone increases sleepiness and which region releases this hormone?
 - The pineal gland releases melatonin
- Stages of sleep
 - Know the different stages and how they differ from each other when assessed by EEG
 - Stages 1 and 2 are light sleep, have theta waves
 - Stage 2 has sleep spindles and K-complexes
 - Stages 3 and 4 are slow-wave sleep, have delta waves
 - REM sleep is vivid dreaming, muscle atonia, waves are smaller
- Pattern of sleep stages
 - How do people “normally” progress through the different stages of sleep?
 - Fall asleep by going through them in sequential order
 - Go back and forth (3 to 4 to 3 to 2 and so on) with REM becoming predominant later on
- Brain mechanisms
 - What brain regions and neural systems underly the different sleep stages?
 -
- Sleep disorders
 - What are the two different sleep disorders we talked about in class?
 - Sleep apnea and narcolepsy
 - Name what causes these disorders and what treatments are available.

- Sleep apnea: tongue falls back leads to low oxygen and is sometimes correlated with lost neurons
 - Treatments = low weight, no alcohol, CPAP (continuous positive airway pressure), surgery
 - Narcolepsy: low hypocretin, which is important for maintaining organized sleep
 - Treatments = stimulants for sleep attacks, antidepressants for REM issues
- Sleep theories
 - What are different theories that explain the function of sleep?
 - Energy conservation
 - Niche adaptation (natural selection emphasizes certain circadian rhythms)
 - Restoration of body
 - Less sleep means higher disease rates and maybe even death
 - Know what is meant by sleep is necessary to restore mind.
 - Sleep deprivation leads to impaired concentration, performance on cognitive tasks, and more
 - Also, more hallucinations
 - What is evidence that shows sleep aids memory?
 - Level of brain activity during sleep correlates with memory improvement
 - More sleep spindles mean more information being exchanged between thalamus and cortex, means better learning
 - We have better recall after sleeping